Jihoon Lee

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RESEARCH INTERESTS

I study how wireless networks can support reliable coordination and intelligent services. My research spans vehicle-to-everything communication, 6G beam management, and semantic communication.

Focus: V2X and distributed control; over-the-air computation; position-aided beam management; semantic communication; AI-enabled radio access networks.

EDUCATION

Ph.D. Program, Electrical & Electronic Engineering

Yonsei University · Mar 2021 - Present

RAMO Laboratory · Advisor: Prof. Seong-Lyun Kim

B.S., Electrical & Electronic Engineering

Yonsei University · Mar 2014 - Feb 2021

Seoul, South Korea

INTERNATIONAL PUBLICATIONS

[1] Enabling Distributed Control of Vehicle Platooning via Over-the-Air Consensus

Jihoon Lee, Yonghoon Jang, Hansol Kim, Seong-Lyun Kim, and Seung-Woo Ko

IEEE Transactions on Wireless Communications, 2024. Vol. 23, No. 11, pp. 17205-17221.

https://doi.org/10.1109/TWC.2024.3451703

[2] Over-the-Air Consensus for Distributed Vehicle Platooning Control

Jihoon Lee, Yonghoon Jang, Hansol Kim, Seong-Lyun Kim, and Seung-Woo Ko

IEEE International Conference on Communications (ICC), 2023. Rome, Italy · pp. 5965-5971.

https://doi.org/10.1109/ICC45041.2023.10279484

[3] Stochastic Geometry Analysis of Position aided Beam Management

Jihoon Lee, Seung-Woo Ko, and Seong-Lyun Kim

ICTC Workshop on Post MIMO, 2023. Jeju, South Korea.

Conference paper

TECHNICAL SKILLS & LANGUAGES

MATLAB, Python, System-level simulation, Stochastic geometry, Resource optimization

Korean: Native · English: Fluent

DOMESTIC CONFERENCES

English translations of Korean titles.

2025 · A Token-Based Approach to Highly Reliable Semantic Communication

Jihoon Lee et al. · KICS Winter Conference · Oral presentation

2023 · An Over-the-Air Consensus Algorithm for Distributed Vehicle Platooning Control

Jihoon Lee, Seung-Woo Ko, and Seong-Lyun Kim · JCCI · Oral presentation

2021 · Reinforcement-Learning-Based Resource Reallocation Control in Vehicular Networks

Jihoon Lee et al. · KICS Conference Proceedings · Conference paper

2021 · Position-Aware MAC Design for mmWave Vehicular Communications

Jihoon Lee, Seungeun Oh, Jinhyuk Choi, and Seong-Lyun Kim · JCCI · Oral presentation

2020 · Single-Channel Sharing in Massive IoT Using Zigbee

Jihoon Lee et al. · KIEES Summer Conference · Poster presentation

RESEARCH PROJECTS

Development of a 5G-A vRAN Research Platform

Apr 2024 - Present · An integrated research platform that standardizes RU, DU, and CU interfaces and supports software-controlled, virtualized Open RAN systems.

Fundamental Technologies for Semantic Communication

Dec 2024 - Apr 2025 · Research on token-based SLM-LLM interaction to improve communication efficiency and inference accuracy.

Sub-THz Augmented Routing and Transmission for 6G

Dec 2021 - Dec 2023 · Research on routing and transmission foundations for next-generation sub-THz communication systems.

High-Accuracy Positioning-Enabled MIMO for 5G-V2X

Sep 2020 - Dec 2021 · A hardware/software testbed project for position-aided MIMO transmission and networking in 5G vehicle-to-everything services.

PATENTS

Selected Korean patents. Titles shown in English translation.

Apparatus and Method for Distributed-Consensus-Based Vehicle Platooning Control

KR 10-2023-0009373 · Application · Jihoon Lee and Seong-Lyun Kim

Method and Apparatus for Empty-Bottle Classification and Quantity Estimation Using Multi-Camera Sequential Machine Vision

KR 10-2613465 · Granted · Co-inventor

Method and Apparatus for Position-Information-Based Vehicular Communication

KR 10-2425933 · Granted · Seungeun Oh, Jihoon Lee, and Seong-Lyun Kim

Method and Apparatus for Resource Reallocation Control in Vehicular Networks

KR 10-2433577 · Granted · Seungeun Oh, Jihoon Lee, and Seong-Lyun Kim